

Deep Learning on jovian decametric emissions

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Direction Informatique de l'Observatoire

Semi-Hack-a-thon ASOV – 25 Mars 2025

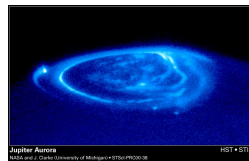


Observatoire
de Paris



Jovian decametric emissions

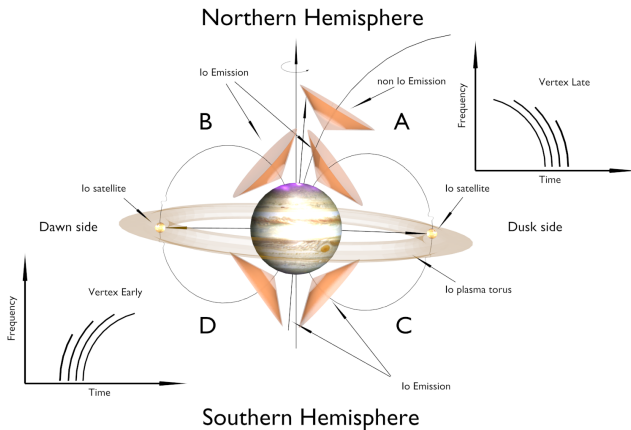
Observed since the 50's, Jovian decametric emissions are produced by the cyclotron-maser instability, due to solar wind or by Jupiter-Io electrodynamic interaction.



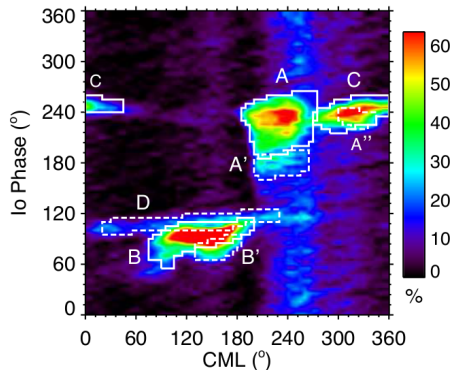
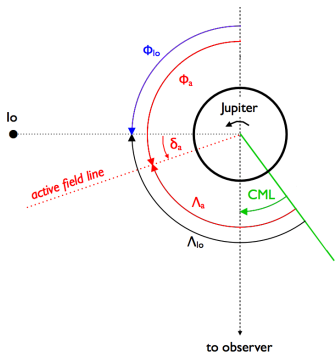
Many timescales:

- 1ms-1s – S-bursts
- 1min-1h – arcs in the time-frequency plane
- ~ 10 h – planetary rotation
- 1d-1m – interplanetary shocks and response to solar wind

Jovian decametric emissions



Jovian decametric emissions



Probability of occurrence wrt. Jupiter rotation and Io ephemeris

Nançay Decameter Array

Built in the 70's

Numerical data since 1990

10 - 100 MHz

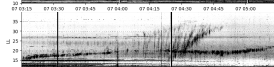
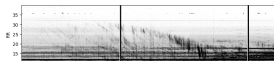
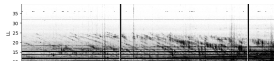
Daily observations of the Sun and Jupiter



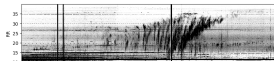
Receiver	# channels	Resolution (ms x kHz)	Data size (8 hours)
Routine	2	500 x 75	22 MB
New Routine	4	500 x 49	568 MB
JunoN	4	2.6 x 3.05	2.9 TB
		83.2 x 12.2	22.6 GB

Observations with the NDA

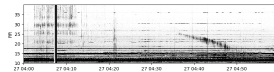
J990807



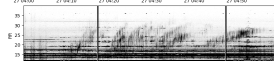
J950509



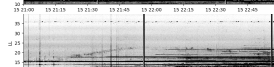
Typical Io decametric emissions



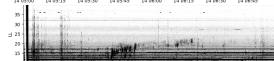
J970627



J001115



J041114

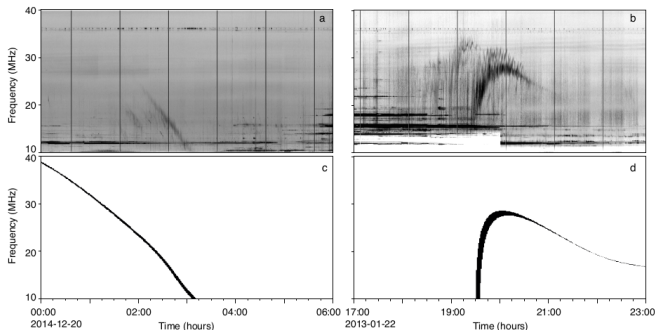


J001027

Typical non-Io decametric emissions

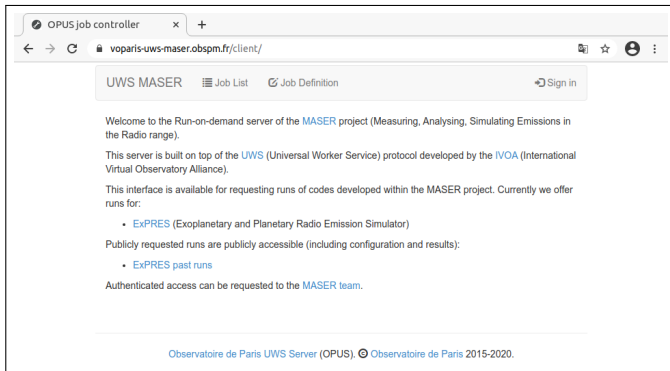
Simulations

ExPRES: an Exoplanetary and Planetary Radio Emissions Simulator (C. K. Louis et al)



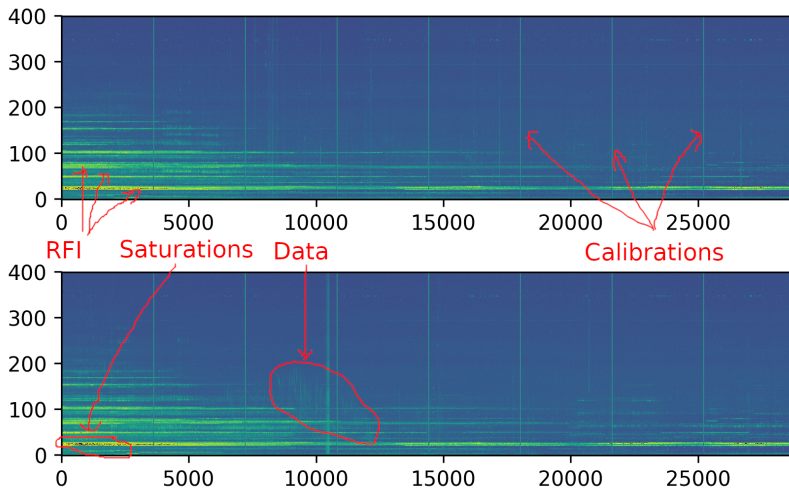
<https://maser.lesia.obspm.fr/task-2-modeling-tools/expres/>

ExPRES : run on demand



<https://voparis-uws-maser.obspm.fr/>

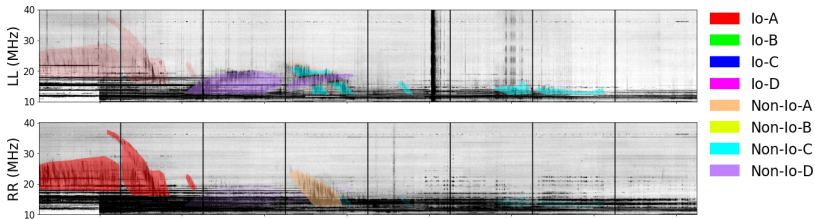
Data exploration



Catalog

Marques et al (2017) built a catalog of 26 years of jovian decametric emissions.

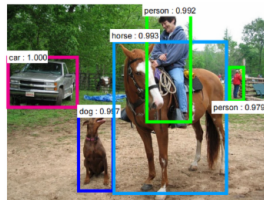
8163 observations (~ 8 hours), 11463 emissions (type, polarization, time range, freq. range, polygon)



Types of classification problem



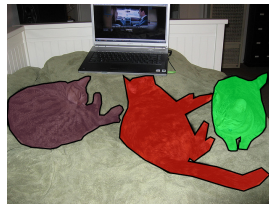
Classification



Localization



Semantic segmentation



Instance segmentation

Semantic segmentation: data preparation

Cut samples of 7680 s into observations.

Clean calibrations, background.

Reduce (mean) to samples of size 384 (time) $\times$ 384 (freq) $\times$ n
(channels = LH, RH, frequencies, ephemeris data)

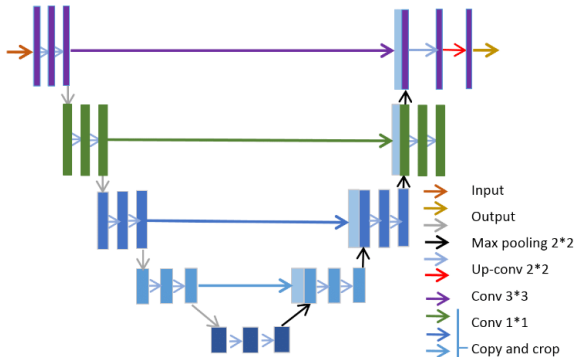
Output = segmentation into 9 types (nothing, Io-A,B,C,D and Non-Io-A,B,C,D)

220750 samples in training set, 7990 samples in validation set

Semantic segmentation: convolutional Network

Unet architecture, with EfficientNetB3 as encoder and a simple convolutional decoder.

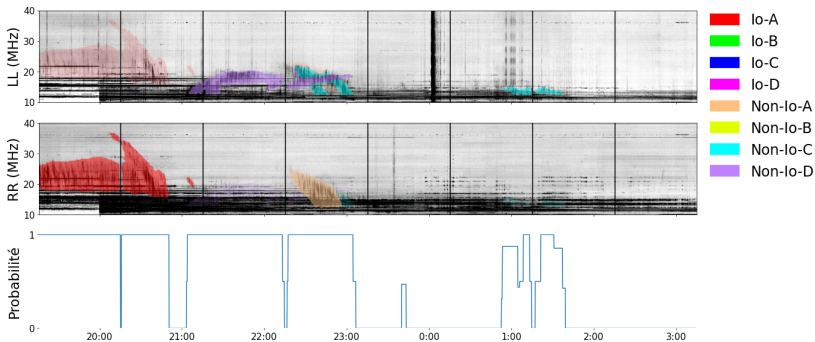
75 M parameters.



Semantic segmentation: results

Detection of emission : $> 97\%$ success

Segmentation : $> 60\%$ IOU.



Visualization tools

Visualization of the catalog and neural network predictions:

<https://voparis-minerva-jupiter.obspm.fr/>

NB: undergoing a redesign, should be OK in june

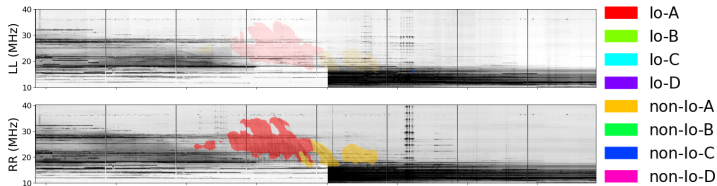
Probability check for an emission:

<https://jupiter-probability-tool.obspm.fr/>

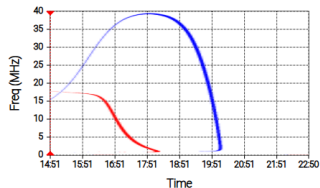
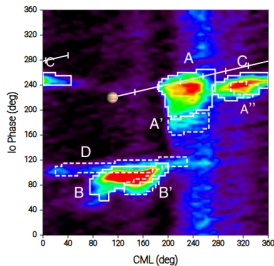
- Check if Io emission is probable
- Compare with ExPRES simulation and observation

Visualization tools

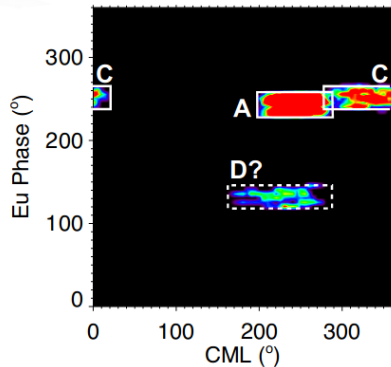
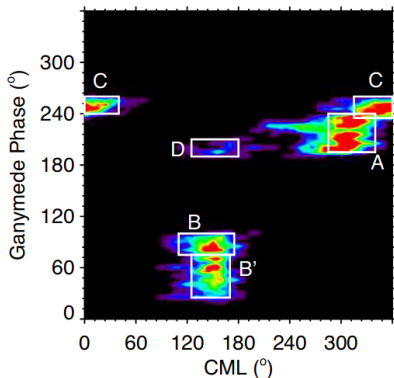
Prediction of Febryuary 14th 2025



View of Jupiter Probability tool of the same day



Perspectives



Apply to emissions due to Ganymede and Europa.

Perspectives

- Publish new version of portal (with multiple neural networks, probability and expres views integrated, statistics and more!)
- Generalize to New routine and Nenufar observations
- test a two-step pipeline with first localisation of emission at full temporal resolution, then a identify the emission types
- Integrate in Extract framework

Thank you!